

Why small tanh slopes make readout gradient descent slow

A finite-network eigenvalue bound, its Fourier explanation, and the resulting training delay

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An accurate readout can exist and still take gradient descent an impractical number of steps to find. For a tanh network with fixed centers, we explain how the common slope γ affects this delay in four steps: express GD through kernel eigenvalue ratios; separate the kernel's explicit gamma dependence from its finite-grid corrections; use a Fourier identity to bound the new ratios; and calculate the resulting training time for the target.

At $\gamma = 4$, a computed readout achieves relative error 5.85×10^{-4} , yet the spectral calculation predicts approximately 10^{12} ordinary readout GD steps to reach even 10^{-2} . At $\gamma = 64$, that step count falls to about 1.83×10^4 . The theorem below accounts for large increases in selected eigenvalue ratios while retaining the boundary and grid corrections. It does not assume that every ratio improves for every geometry.

1. With frozen geometry, GD repeatedly applies the same matrix to the residual

We have m training inputs $x_1, \dots, x_m$ and W tanh neurons, including any halo neurons whose centers lie outside the training interval. The centers c_j and common slope $\gamma > 0$ are fixed. Define the normalized feature matrix $B_\gamma \in \mathbb{R}^{m \times (W+1)}$ by

$$(B_\gamma)_{aj} = \frac{\tanh(\gamma(x_a - c_j))}{\sqrt{m}}, \quad (B_\gamma)_{a0} = \frac{1}{\sqrt{m}}.$$

Column zero is the constant feature. The trainable vector v contains the readout coefficients and output bias. With target values $y_a = f(x_a)/\sqrt{m}$, the loss

$$L(v) = \frac{1}{2} \|B_\gamma v - y\|^2$$

is half the mean squared training error. Write $r_n = B_\gamma v_n - y$ for the normalized residual after n updates. Its readout gradient is $B_\gamma^\top r_n$, so

$$\begin{aligned} v_{n+1} &= v_n - \eta_\gamma B_\gamma^\top r_n, \\ r_{n+1} &= B_\gamma(v_n - \eta_\gamma B_\gamma^\top r_n) - y \\ &= (I - \eta_\gamma K_\gamma) r_n, \quad K_\gamma := B_\gamma B_\gamma^\top. \end{aligned} \tag{1}$$

Starting from zero readout gives $r_0 = -y$, and therefore $r_n = -(I - \eta_\gamma K_\gamma)^n y$. Let $\lambda_1(\gamma) \geq \dots \geq \lambda_m(\gamma) \geq 0$ be the eigenvalues of K_γ , with orthonormal eigenvectors $u_i(\gamma)$. Choose the learning rate

$$\eta_\gamma = \frac{1}{2\lambda_1(\gamma)}.$$

Then each residual component evolves independently:

$$u_i(\gamma)^\top r_n = -u_i(\gamma)^\top y \left(1 - \frac{1}{2} \frac{\lambda_i(\gamma)}{\lambda_1(\gamma)}\right)^n. \tag{2}$$

If λ_i/λ_1 is tiny, the factor in parentheses is almost one and that component decreases slowly. Scaling the entire kernel by one constant leaves this factor unchanged; changing the ratios changes the convergence rates.

The target matters as well. Define $p_i(\gamma) = |u_i(\gamma)^\top y|^2 / \|y\|^2$, the fraction of its squared norm in eigenvector i . The relative training error is exactly

$$E_\gamma(n)^2 := \frac{\|r_n\|^2}{\|y\|^2} = \sum_{i=1}^m p_i(\gamma) \left(1 - \frac{1}{2} \frac{\lambda_i(\gamma)}{\lambda_1(\gamma)}\right)^{2n}. \quad (3)$$

Zero eigenvalues contribute an error that readout training cannot remove. Positive eigenvalues describe components that are representable but may take many steps to learn.

For the mixed-sine target, holding the target weights at $\gamma_0 = 8$ while changing the ratios reproduces the main orders of magnitude in the decrease in step count (Figure 1). Changing only the weights does not.

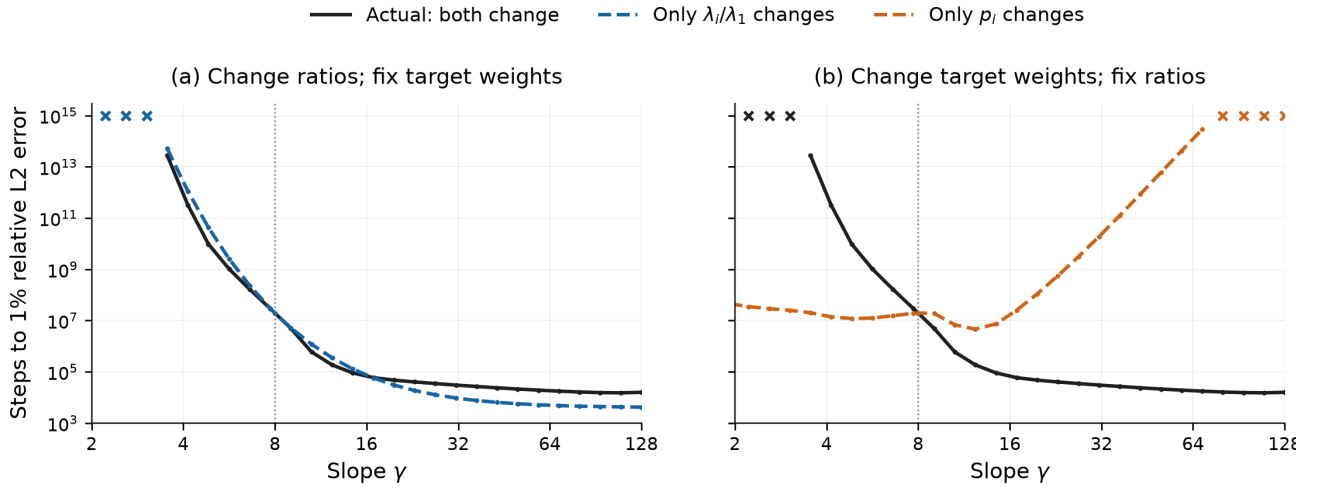


Figure 1. Steps to 1% relative training error from the finite-kernel spectral formula. Left: the full calculation and a calculation that changes the normalized eigenvalues while holding p_i at $\gamma_0 = 8$. Right: the full calculation and the reverse control. Crosses indicate that 1% was not reached by 10^{15} steps. Components are paired by descending eigenvalue rank, not by following individual eigenvectors. Shading shows sensitivity to numerically unresolved directions; it is not a statistical confidence interval. Appendix E specifies the geometry and numerical controls.

2. The matrix decomposition makes gamma's contribution explicit

The kernel entries are sums of feature products:

$$(K_\gamma)_{ab} = \frac{1}{m} \left[1 + \sum_{j=1}^W g_{ab}(c_j) \right], \quad g_{ab}(c) = \tanh(\gamma(x_a - c)) \tanh(\gamma(x_b - c)).$$

We now consider a continuous version of this matrix, denoted K_γ^{int} , by replacing the finite feature inner product with an integral inner product over functions of the center coordinate c . The data locations x_a, x_b remain fixed.

Assume uniformly spaced centers with spacing h . Integrate over their cells, from $c_- = c_1 - h/2$ to $c_+ = c_W + h/2$, an interval of length Wh :

$$(K_\gamma^{\text{int}})_{ab} = \frac{1}{m} \left[1 + \frac{1}{h} \int_{c_-}^{c_+} g_{ab}(c) dc \right], \quad E_\gamma^{\text{grid}} := K_\gamma - K_\gamma^{\text{int}}. \quad (4)$$

The density $1/h$ matches the scale of the neuron sum; E^{grid} records the exact difference.

Write $g_{ab} = 1 - (1 - g_{ab})$. Integrating the constant gives Wh , and the remaining whole-line integral has a closed form:

$$\int_{\mathbb{R}} [1 - g_{ab}(c)] dc = 2d_{ab} \coth(\gamma d_{ab}), \quad d_{ab} := x_a - x_b.$$

At $d_{ab} = 0$, its continuous value is $2/\gamma$. The finite interval omits two tails. Define their contribution by

$$(E_{\gamma}^{\text{halo}})_{ab} = \frac{1}{hm} \int_{\mathbb{R} \setminus [c_-, c_+]} [1 - g_{ab}(c)] dc.$$

The resulting decomposition is exact:

$$K_{\gamma} = \underbrace{\frac{W+1}{m} \mathbf{1}\mathbf{1}^{\top} - \frac{2}{hm} [d_{ab} \coth(\gamma d_{ab})]_{a,b}}_{K_{\gamma}^{(0)}} + \underbrace{E_{\gamma}^{\text{halo}} + E_{\gamma}^{\text{grid}}}_{E_{\gamma}}. \quad (5)$$

The explicit matrix $K_{\gamma}^{(0)}$ has two terms: a constant rank-one matrix and a matrix depending only on the separation $x_a - x_b$. Translating both inputs by the same amount leaves this separation, and hence the entry, unchanged. On a uniformly spaced sample grid,

$$(K_{\gamma}^{(0)})_{a+1,b+1} = (K_{\gamma}^{(0)})_{ab}.$$

Its entries are therefore constant along each diagonal: this is the Toeplitz structure. The bracketed term in (5) is understood by continuity at $a = b$, where its value is $1/\gamma$.

Equation (5) separates this translation-invariant part from E_{γ} , which retains the effects of the finite center interval and discrete centers. Those corrections can depend on the two input positions separately. The Fourier calculation below uses the dependence on separation in $K_{\gamma}^{(0)}$; it does not assume that Fourier waves are eigenvectors of the finite matrix.

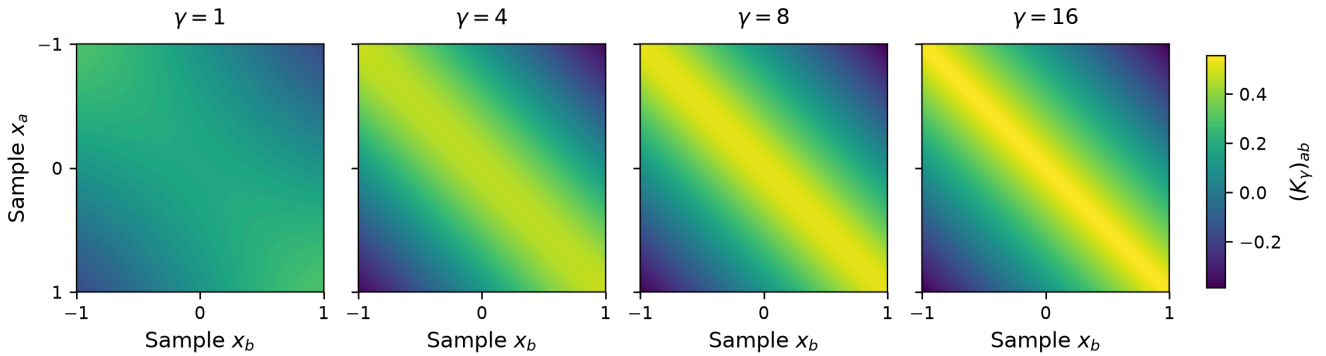


Figure 2. Actual matrices $K_{\gamma} = B_{\gamma} B_{\gamma}^{\top}$ at $\gamma = 1, 4, 8, 16$, with identical centers, samples, and a shared color scale. The diagonal bands reflect dependence on the separation $x_a - x_b$. Departures from constant values along a diagonal are retained in E_{γ} ; these plots show the full finite matrices, not just their Toeplitz part.

3. The Fourier change gives a finite lower bound on the eigenvalue ratio

Compare two slopes $0 < \gamma_a < \gamma_b$, with samples, centers, and readout coordinates held fixed. Tanh is an average of a step:

$$\rho_\gamma(x) = \frac{\gamma}{2} \text{sech}^2(\gamma x), \quad (\rho_\gamma * \text{sgn})(x) = 2 \int_{-\infty}^x \rho_\gamma(t) dt - 1 = \tanh(\gamma x).$$

With Fourier convention $\hat{f}(\omega) = \int f(x) e^{-i\omega x} dx$, convolution multiplies the transform by

$$M_\gamma(\omega) = \hat{\rho}_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_\gamma(0) = 1. \quad (6)$$

For $|\omega| \ll \gamma$, this multiplier is close to one; for $|\omega| \gg \gamma$, it is approximately $2ze^{-z}$. This suppression appears in the kernel itself. Write $k_\gamma(x, x')$ for the feature-product sum before sampling, including the bias, and k_{step} for the same sum with $\tanh$ replaced by sign functions. Averaging acts on both inputs:

$$\widehat{k}_\gamma(\omega, \omega') = M_\gamma(\omega) M_\gamma(\omega') \widehat{k}_{\text{step}}(\omega, \omega').$$

Small gamma therefore reduces high-frequency variation in the kernel as either input changes. Evaluating the resulting function at (x_a, x_b) gives the matrix entries, with normalization $(K_\gamma)_{ab} = k_\gamma(x_a, x_b)/m$.

We now connect this change to an eigenvalue. Take a unit eigenvector u of the old kernel: $K_{\gamma_a} u = \lambda_i(\gamma_a) u$. Then $u^\top K_{\gamma_a} u = \lambda_i(\gamma_a)$. We calculate its change using $\Delta K^{(0)} := K_{\gamma_b}^{(0)} - K_{\gamma_a}^{(0)}$.

The derivation has three steps.

1. The constant rank-one term in (5) cancels. The change in the separation-dependent term is

$$D(d) = 2[d \coth(\gamma_a d) - d \coth(\gamma_b d)], \quad (\Delta K^{(0)})_{ab} = \frac{D(x_a - x_b)}{hm}.$$

Twice differentiating the center integral gives

$$D'' = 4(\rho_{\gamma_a} * \rho_{\gamma_a} - \rho_{\gamma_b} * \rho_{\gamma_b}) \implies \widehat{D}(\omega) = 4 \frac{M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2}{\omega^2}.$$

The implication uses $\widehat{D}'' = -\omega^2 \widehat{D}$ and the convolution rule. Appendix B supplies the integration by parts and the value at zero.

2. Fourier inversion gives the change of each matrix entry:

$$(\Delta K^{(0)})_{ab} = \frac{2}{\pi h m} \int_{\mathbb{R}} \frac{M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2}{\omega^2} e^{i\omega(x_a - x_b)} d\omega.$$

3. Expand $u^\top \Delta K^{(0)} u = \sum_{a,b} u_a u_b (\Delta K^{(0)})_{ab}$. The double sum of exponentials becomes $|\sum_a u_a e^{-i\omega x_a}|^2$, yielding

$$\boxed{u^\top \Delta K^{(0)} u = \frac{2}{\pi h m} \int_{\mathbb{R}} \frac{M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2}{\omega^2} \left| \sum_{a=1}^m u_a e^{-i\omega x_a} \right|^2 d\omega.} \quad (7)$$

The identity holds for every real u . For our eigenvector, the last squared factor is its squared Fourier amplitude on the training grid. Holding that eigenvector fixed, all gamma dependence is in $M_{\gamma_b}^2 - M_{\gamma_a}^2$. Increasing gamma makes this difference nonnegative, so (7) adds a nonnegative amount to the old eigenvalue's quadratic form:

$$u^\top K_{\gamma_b} u = \lambda_i(\gamma_a) + \underbrace{u^\top \Delta K^{(0)} u}_{\text{Fourier integral in (7)}} + u^\top (E_{\gamma_b} - E_{\gamma_a}) u.$$

This explains how the changes can be very different across the spectrum. Where both multipliers are already near one, their difference is small. Where the old multiplier was exponentially small, the new multiplier can be many times larger. An eigenvector whose quadratic form depends on those frequencies can therefore gain much more relative to its old value. The leading eigenvalue need not be constant, but in our $8 \rightarrow 16$ comparison it grows by only about 1.6%, while λ_{32} grows by about 1,300 times.

The new eigenvectors may change, so the displayed quadratic form is not automatically the new i -th eigenvalue. To bound that eigenvalue, apply (7) to every vector in the span of the first i old eigenvectors. Let U_i contain those eigenvectors as columns. On this whole space, the new kernel is

$$U_i^\top K_{\gamma_b} U_i = C_i + R_i.$$

Here C_i contains the old eigenvalues plus the Fourier increase; R_i contains the change in the boundary and grid corrections.

Theorem 1 (finite-network ratio bound). Assume the geometry of Section 2 and $\lambda_i(\gamma_a) > 0$. With $U_i \in \mathbb{R}^{m \times i}$ as above, define

$$\begin{aligned} C_i &= \text{diag}(\lambda_1(\gamma_a), \dots, \lambda_i(\gamma_a)) + U_i^\top \Delta K^{(0)} U_i, \\ R_i &= U_i^\top (E_{\gamma_b} - E_{\gamma_a}) U_i, \\ \varepsilon_i &= \sup_{z \in \mathbb{R}^i \setminus \{0\}} \frac{|z^\top R_i z|}{z^\top C_i z}. \end{aligned} \tag{8}$$

The number ε_i bounds the correction relative to C_i for every vector in this space. For any $L_b \geq \lambda_1(\gamma_b)$,

$$\frac{\lambda_i(\gamma_b)}{\lambda_1(\gamma_b)} \geq \frac{[1 - \varepsilon_i]_+ \lambda_{\min}(C_i)}{L_b}, \quad [t]_+ := \max(t, 0). \tag{9}$$

To see why, minimize the new quadratic form over unit vectors in this old i -dimensional space. The minimum is $\lambda_{\min}(C_i + R_i)$. The new i -th eigenvalue is at least this value, because the min–max theorem chooses the best such space:

$$\lambda_i(\gamma_b) \geq \lambda_{\min}(U_i^\top K_{\gamma_b} U_i) = \lambda_{\min}(C_i + R_i) \geq [1 - \varepsilon_i]_+ \lambda_{\min}(C_i).$$

Division by L_b gives (9). Thus (7) supplies the increase in (8); minimizing over the old subspace and allowing for the correction gives the ratio bound (9). Improvement is proved when that bound exceeds the starting ratio. Appendix C contains the full proof.

For rank 32 in the default finite network, the evaluated bounds are:

Gamma change	Starting ratio	Lower bound on new ratio	Actual new ratio
8 $\rightarrow$ 16	4.93×10^{-9}	3.40×10^{-6}	6.30×10^{-6}
16 $\rightarrow$ 64	6.30×10^{-6}	3.39×10^{-4}	4.67×10^{-4}

The first comparison establishes a ratio increase of at least about 690 times. These calculations use the maximum absolute row sum of the new kernel for L_b .

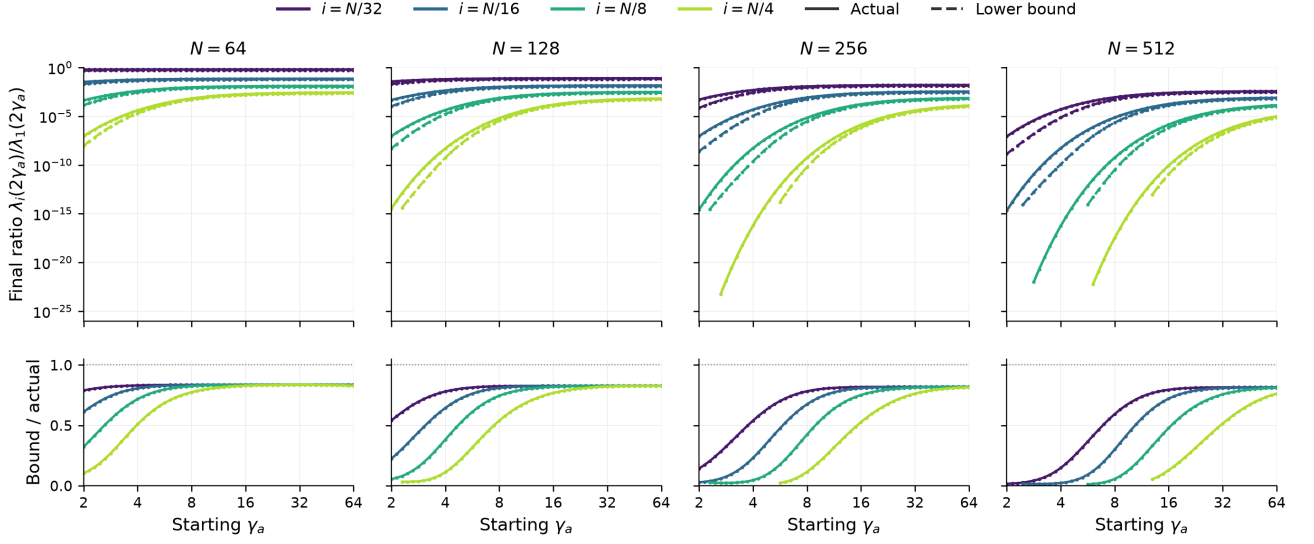


Figure 3. Each comparison doubles γ_a to $2\gamma_a$, at fixed geometry. Top: actual new ratios (solid) and the lower bound (dashed), for four ranks and four widths. Bottom: lower bound divided by the actual ratio, exposing where the bound is loose. Here N counts intervals in $[-1, 1]$, not all neurons: the total tanh widths are $W = 81, 153, 289, 559$. The ranks are $N/32, N/16, N/8, N/4$. Gaps omit numerically unresolved starting subspaces. A positive or fairly tight bound need not exceed the starting ratio; the rank-32 comparison above does.

The bound retains the actual geometry: we evaluate the correction and L_b using the new finite kernel. It establishes these comparisons, rather than a universal ordering of every ratio as gamma increases.

4. Small ratios force a long training delay even when an accurate readout exists

A small eigenvalue matters to this target only if its coefficient in (3) is nonzero. To measure that requirement, choose a ratio cutoff $0 < \rho \leq 1$, and define

$$S_\gamma(\rho) = \left\{ i : 0 < \frac{\lambda_i(\gamma)}{\lambda_1(\gamma)} \leq \rho \right\}, \quad p_\gamma(\rho) = \sum_{i \in S_\gamma(\rho)} p_i(\gamma).$$

The set contains positive eigenvalues only, so these components can be represented by the readout. Their squared residuals add because the eigenvectors are orthonormal. Keeping only this part of (3) gives

$$\begin{aligned} E_\gamma(n)^2 &\geq \sum_{i \in S_\gamma(\rho)} p_i(\gamma) \left(1 - \frac{1}{2} \lambda_i(\gamma) / \lambda_1(\gamma)\right)^{2n} \\ &\geq p_\gamma(\rho) (1 - \rho/2)^{2n}. \end{aligned}$$

Consequently, attaining relative error $\epsilon < \sqrt{p_\gamma(\rho)}$ requires

$$\boxed{n \geq \frac{\log(\sqrt{p_\gamma(\rho)}/\epsilon)}{-\log(1 - \rho/2)} \simeq \frac{2}{\rho} \log \frac{\sqrt{p_\gamma(\rho)}}{\epsilon}.} \quad (10)$$

The approximation uses small ρ ; the preceding bound is exact, with a ceiling for integer steps. More generally, any selected positive-eigenvalue band with ratios at most ρ and known target energy gives the same bound.

For a concrete test, take

$$f(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x)$$

on 263 samples in $[-1, 1]$, with 153 uniformly spaced tanh neurons including the halo. At $\gamma = 4$, a selected band of positive per-step rates $\eta_\gamma \lambda_i \in (10^{-14}, 10^{-11}]$ contains 2.735% of the target’s squared norm. Taking $\rho = 2 \times 10^{-11}$ and that band’s energy in (10) requires at least 2.81×10^{11} steps to reach 1% error. The full sum (3) puts the crossing at approximately 1.008×10^{12} steps.

An explicitly constructed readout at the same $\gamma = 4$ achieves relative error 5.85×10^{-4} , well below 1%. The accurate fit exists, but ordinary GD requires an impractical number of steps to reach it. Reaching a smaller error would require at least as many steps.

Allowing a scalar schedule with every step in the conventional stability range $0 \leq \eta_n \leq 2/\lambda_1$ still requires approximately 7.01×10^{10} steps. Such a schedule can multiply the band’s rates by at most four relative to our choice. Appendix D derives this bound.

To predict the step counts, we evaluate (3) using the center-integral matrix K_γ^{int} from (4), with the same samples, finite integration bounds, and learning rate as the finite-neuron calculation. No convergence rate is fitted, and no trillion-step training run is executed.

γ	Finite-kernel steps to 1%	Integral prediction	Relative prediction error	Explicit readout error
4	1.0076×10^{12}	1.0090×10^{12}	+0.139%	5.85×10^{-4}
8	19,697,563	19,698,115	+0.00280%	2.92×10^{-6}
16	66,209	66,209	0%	1.56×10^{-7}
64	18,272	18,219	-0.290%	5.97×10^{-5}

The prediction error is below 0.3% at all four slopes. The explicit readout errors are attained errors of particular computed coefficients, not assertions about an exact least-squares floor. All four are below the requested tolerance.

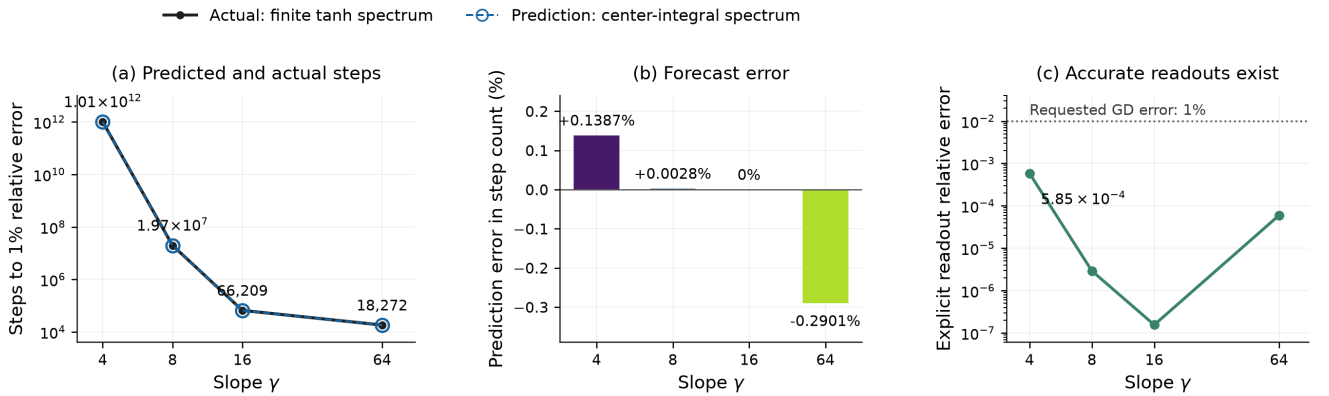


Figure 4. Left: the steps to 1% from the actual finite-tanh spectrum and from the center-integral spectrum. Center: their relative difference. Right: explicitly attained readout errors, with 1% marked for comparison. “Actual” means the spectral calculation for the actual finite matrix, not an executed optimization trajectory. The integral forecast is a separate numerical approximation; it is not obtained by substituting the ratio lower bounds of Theorem 1 into a full-target training-time bound.

Appendix A. Exact center-integral decomposition

Let $c_j = c_1 + (j - 1)h$, and choose $c_- = c_1 - h/2$, $c_+ = c_W + h/2$. The length Wh is essential to the constant $(W + 1)/m$ in (5). Integrating only between the outermost centers would give the wrong constant.

For $d = x_a - x_b \neq 0$, set $u = \gamma(x_a - c)$ and $v = \gamma(x_b - c)$. Then

$$1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v).$$

The difference has a convergent integral, and the log-cosh antiderivative gives

$$\int_{\mathbb{R}} [\tanh(\gamma(x_a - c)) - \tanh(\gamma(x_b - c))] dc = 2d.$$

Multiplication by $\coth(\gamma d)$ proves the whole-line deficit formula. When $d = 0$, the integrand is $\text{sech}^2(\gamma(x_a - c))$, with integral $2/\gamma$.

Splitting the whole-line deficit into the center interval and its complement gives

$$\begin{aligned} (K_{\gamma}^{\text{int}})_{ab} &= \frac{W + 1}{m} - \frac{1}{hm} \int_{c_-}^{c_+} (1 - g_{ab}) dc \\ &= \frac{W + 1}{m} - \frac{2d \coth(\gamma d)}{hm} + (E_{\gamma}^{\text{halo}})_{ab}. \end{aligned}$$

Adding E_{γ}^{grid} proves (5). No small-error assumption is used. Although the halo correction is entrywise nonnegative, it need not be positive semidefinite. Neither is $K_{\gamma}^{(0)}$ asserted to be positive semidefinite. Section 3 proves positivity for its finite increase $\Delta K^{(0)}$.

Appendix B. Fourier identity, including the constant component

B.1. The transform of the averaging density

The cumulative integral of $\rho_{\gamma}(t) = \gamma \text{sech}^2(\gamma t)/2$ is $(1 + \tanh(\gamma t))/2$, proving the convolution identity in the main text. To calculate its transform, substitute $s = e^{2\gamma t}$ and write $\nu = \omega/(2\gamma)$:

$$\hat{\rho}_{\gamma}(\omega) = \int_0^{\infty} \frac{s^{-i\nu}}{(1 + s)^2} ds = \Gamma(1 - i\nu)\Gamma(1 + i\nu) = \frac{\pi\nu}{\sinh(\pi\nu)}.$$

The middle identity is the beta integral; the last uses the gamma reflection formula. Continuity gives the value one at zero, establishing (6).

For the kernel identity in the main text, let $k_{\text{step}}(x, x') = 1 + \sum_j \text{sgn}(x - c_j) \text{sgn}(x' - c_j)$. Averaging in each input gives

$$k_{\gamma}(x, x') = \int_{\mathbb{R}^2} \rho_{\gamma}(x - t) \rho_{\gamma}(x' - s) k_{\text{step}}(t, s) dt ds.$$

The two-input Fourier identity follows by the convolution rule, with transforms interpreted as tempered distributions because the kernels have nondecaying tails. The proof of (7) below instead transforms the integrable difference D , for which ordinary Fourier inversion applies.

For $z > 0$, $\partial_z \log(z/\sinh z) = 1/z - \coth z < 0$, so M_γ increases with gamma. The finite gain $M_{\gamma_b}(\omega)/M_{\gamma_a}(\omega)$ increases with $|\omega|$ when $\gamma_b > \gamma_a$: differentiating its logarithm with respect to $|\omega|$ gives the positive difference between $a \coth(a|\omega|)$ and $b \coth(b|\omega|)$, where $a = \pi/(2\gamma_a) > b = \pi/(2\gamma_b)$. Positivity follows because $z \coth z$ increases for $z > 0$.

B.2. Transforming the change in the center integral

Write $T_\gamma(s) = \tanh(\gamma s)$. Appendix A gives

$$I_\gamma(d) := \int_{\mathbb{R}} [1 - T_\gamma(s+d)T_\gamma(s)]ds = 2d \coth(\gamma d).$$

The first two derivatives of the integrand are integrable, locally uniformly in d , because the derivatives of $\tanh$ decay exponentially. Differentiate twice, then integrate by parts in s :

$$\begin{aligned} I_\gamma''(d) &= - \int_{\mathbb{R}} T_\gamma''(s+d)T_\gamma(s)ds \\ &= \int_{\mathbb{R}} T_\gamma'(s+d)T_\gamma'(s)ds \\ &= 4(\rho_\gamma * \rho_\gamma)(d). \end{aligned} \tag{B1}$$

The boundary term vanishes; the last equality uses $T_\gamma' = 2\rho_\gamma$ and the evenness of ρ_γ .

The difference $D = I_{\gamma_a} - I_{\gamma_b}$, unlike either individual I_γ , decays at infinity. Its explicit formula shows that D, D' , and D'' are integrable. The ordinary Fourier derivative and convolution rules therefore give

$$-\omega^2 \widehat{D}(\omega) = 4(M_{\gamma_a}(\omega)^2 - M_{\gamma_b}(\omega)^2), \quad \widehat{D}(\omega) = 4 \frac{M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2}{\omega^2}. \tag{B2}$$

At zero, expand $M_\gamma^2 = 1 - \pi^2\omega^2/(12\gamma^2) + O(\omega^4)$. The continuous value is

$$\widehat{D}(0) = \frac{\pi^2}{3}(\gamma_a^{-2} - \gamma_b^{-2}).$$

Thus $\widehat{D}$ is integrable, and Fourier inversion gives

$$D(d) = \frac{2}{\pi} \int_{\mathbb{R}} \frac{M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2}{\omega^2} e^{i\omega d} d\omega.$$

B.3. From matrix entries to equation (7)

Since $(\Delta K^{(0)})_{ab} = D(x_a - x_b)/(hm)$, substitute this inverse transform into the finite sum for $u^\top \Delta K^{(0)} u$. The remaining factor is

$$\sum_{a,b} u_a u_b e^{i\omega(x_a - x_b)} = \left(\sum_a u_a e^{i\omega x_a} \right) \left(\sum_b u_b e^{-i\omega x_b} \right) = \left| \sum_a u_a e^{-i\omega x_a} \right|^2.$$

This proves (7) for every real u , including vectors with nonzero sum. Since $M_{\gamma_b} \geq M_{\gamma_a}$, the integrand is nonnegative and $\Delta K^{(0)}$ is positive semidefinite. The proof uses the ordinary transform of the decaying difference D , so no distributional transform or unspecified constant is needed.

Appendix C. Proof and evaluation of the ratio bound

The old top- i eigenvectors satisfy

$$U_i^\top K_{\gamma_a} U_i = \text{diag}(\lambda_1(\gamma_a), \dots, \lambda_i(\gamma_a)).$$

The exact decomposition gives $U_i^\top K_{\gamma_b} U_i = C_i + R_i$. Because $\lambda_i(\gamma_a) > 0$ and $\Delta K^{(0)} \succeq 0$, C_i is positive definite. The relative correction is equivalently

$$\varepsilon_i = \|C_i^{-1/2} R_i C_i^{-1/2}\|_2.$$

For every z , its definition gives $z^\top R_i z \geq -\varepsilon_i z^\top C_i z$. If $\varepsilon_i < 1$, then $C_i + R_i \succeq (1 - \varepsilon_i)C_i$. If $\varepsilon_i \geq 1$, use $C_i + R_i = U_i^\top K_{\gamma_b} U_i \succeq 0$. Therefore

$$\lambda_{\min}(C_i + R_i) \geq [1 - \varepsilon_i]_+ \lambda_{\min}(C_i).$$

Courant–Fischer, with eigenvalues ordered from largest to smallest, states

$$\lambda_i(K_{\gamma_b}) = \max_{\dim V=i} \min_{u \in V, \|u\|=1} u^\top K_{\gamma_b} u.$$

Choosing $V = \text{range}(U_i)$ proves the numerator bound in (9). Since the numerator is nonnegative, replacing $\lambda_1(\gamma_b)$ by an upper bound L_b preserves the lower bound on the ratio.

Two choices require no new eigendecomposition: $L_b = W + 1$, since $\text{tr } K_{\gamma_b} \leq W + 1$, and the tighter bound used in the figures,

$$L_b = \max_a \sum_{b=1}^m |(K_{\gamma_b})_{ab}|.$$

Absolute values are required: kernel entries can be negative. The same proof works for any orthonormal trial basis U_i if the old diagonal block is replaced by $U_i^\top K_{\gamma_a} U_i$, provided the resulting C_i is positive definite. This formulation also avoids treating numerical singular vectors as exact eigenvectors.

The computed ε_i retains both halo and grid effects relative to the matrix on the tested space. A small global error compared with λ_1 would not be sufficient: it could still be larger than the small eigenvalues under study. We therefore do not discard the corrections using a global “small error” claim.

Nor does the theorem assume that the larger-gamma normalized spectrum improves at every rank. For example, in the $N = 128$, $8 \rightarrow 16$ comparison, the rank-4 lower bound is about 82.4% of the actual new ratio but only 97.3% of the old ratio. It is fairly tight without proving improvement. At rank 32, the lower bound is about 54.0% of the actual new ratio and about 690 times the old ratio, so it does prove a large increase. The fixed trial space and the upper bound on the denominator both contribute conservatism.

Appendix D. Necessary steps and the role of the target

For a positive eigenvalue, its eigenvector lies in the representable span:

$$u_i = B_\gamma \left(\frac{B_\gamma^\top u_i}{\lambda_i} \right).$$

Thus a target component in a positive-eigenvalue band is exactly representable on the training grid. This says nothing about how large the required coefficients are or how quickly GD reaches them.

Taking the orthogonal projection of the residual onto a selected band S with $0 < \lambda_i/\lambda_1 \leq \rho$ gives

$$\frac{\|\sum_{i \in S} u_i u_i^\top r_n\|^2}{\|y\|^2} = \sum_{i \in S} p_i (1 - \lambda_i/(2\lambda_1))^{2n} \geq \left(\sum_{i \in S} p_i \right) (1 - \rho/2)^{2n}.$$

The total error is at least this projected error. Taking square roots, requiring it to be at most ϵ , and solving for n proves (10). A nonzero nullspace component instead creates an irreducible floor; it is deliberately excluded from this necessary-delay example.

For the scalar schedule claim, use the same fixed kernel with $0 \leq \eta_t \leq 2/\lambda_1$. If a band obeys $\lambda_i/(2\lambda_1) \leq b$, then $\eta_t \lambda_i \leq 4b$. For $4b < 1$, all its residual factors obey $1 - \eta_t \lambda_i \geq 1 - 4b > 0$. Hence

$$E(n)^2 \geq \sum_{i \in S} p_i \prod_{t=0}^{n-1} (1 - \eta_t \lambda_i)^2 \geq p(1 - 4b)^{2n}, \quad p := \sum_{i \in S} p_i.$$

At $\gamma = 4$, $b = 10^{-11}$ and $p = 0.02735465123$ give approximately 7.01×10^{10} necessary steps to 1%. This applies to scalar GD schedules within the stated per-step stability range, not to all possible first-order algorithms.

Finally, the direction of the ratio inequality matters. If lower ratio bounds $\ell_i \leq \lambda_i/\lambda_1$ are available, then

$$p_i (1 - \lambda_i/(2\lambda_1))^{2n} \leq p_i (1 - \ell_i/2)^{2n}.$$

These can give an upper bound on error, provided the other target components are included. A bound at a few ranks alone cannot predict the full target's hitting time. That is why the step-count figure uses the full integral spectrum and target weights rather than treating Theorem 1 as a full learning-curve formula.

Appendix E. Experiments, predictions, and numerical resolution

E.1. Geometry and target

The default calculation uses $N = 128$ equal intervals on $[-1, 1]$, with spacing $h = 2/N = 1/64$. There are $N + 1 = 129$ interior centers and $H = \lceil \sqrt{N} \rceil = 12$ halo centers on each side. Thus $W = N + 1 + 2H = 153$ tanh neurons, plus the constant feature. The center-cell interval is $[-1.1953125, 1.1953125]$. The $m = 263$ training inputs are equally spaced on $[-1, 1]$, including endpoints. The target is the mixed sine in Section 4, and all reported errors are relative norms on these training inputs.

For the width comparisons:

Interior intervals N	Total tanh neurons W	Samples m	Halo per side	Tested ranks
64	81	131	8	2, 4, 8, 16
128	153	263	12	4, 8, 16, 32
256	289	521	16	8, 16, 32, 64
512	559	1031	23	16, 32, 64, 128

Each gamma comparison fixes its samples and centers. Across widths, spacing and physical halo extent change with this rule; the columns should not be read as changing only the number of neurons while preserving all other geometric lengths. The eigenvalue-ratio figure is independent of the target. The training-time and target-weight figures concern the stated mixed sine.

E.2. How the integral forecast is calculated

We approximate the center integral in (4) using Gauss–Legendre quadrature within each center cell. If t_ℓ and $w_\ell > 0$ are its nodes and weights over the whole center interval, form a weighted feature matrix with constant column $1/\sqrt{m}$ and columns

$$A_{a\ell} = \sqrt{\frac{w_\ell}{hm}} \tanh(\gamma(x_a - t_\ell)).$$

Then $AA^\top$ approximates K_γ^{int} . This keeps the original training samples and the finite halo interval. It approximates the center sum; it is not a periodic Fourier construction.

From its singular values and left singular vectors, calculate eigenvalues $\tilde{\lambda}_i$ and target weights $\tilde{p}_i$. With the same $\eta_\gamma = 1/(2\lambda_1(K_\gamma))$ used for the finite matrix, the predicted curve is

$$\tilde{E}_\gamma(n)^2 = \sum_i \tilde{p}_i (1 - \eta_\gamma \tilde{\lambda}_i)^{2n},$$

including any target norm outside the computed column span. The first integer crossing of 1% is found by bracketing and binary search. Tiny decay rates are evaluated using $\exp(2n \log 1p(-\eta_\gamma \tilde{\lambda}_i))$, rather than subtracting them from one and repeatedly multiplying.

Eight and sixteen quadrature points per cell give nearly identical step predictions. At $\gamma = 4$, their predictions differ by 57 steps out of approximately 10^{12} ; at the other three slopes their integer predictions agree. The agreement with the finite matrix in Figure 4 is a measured numerical result, not a certified enclosure of all rounding and quadrature errors.

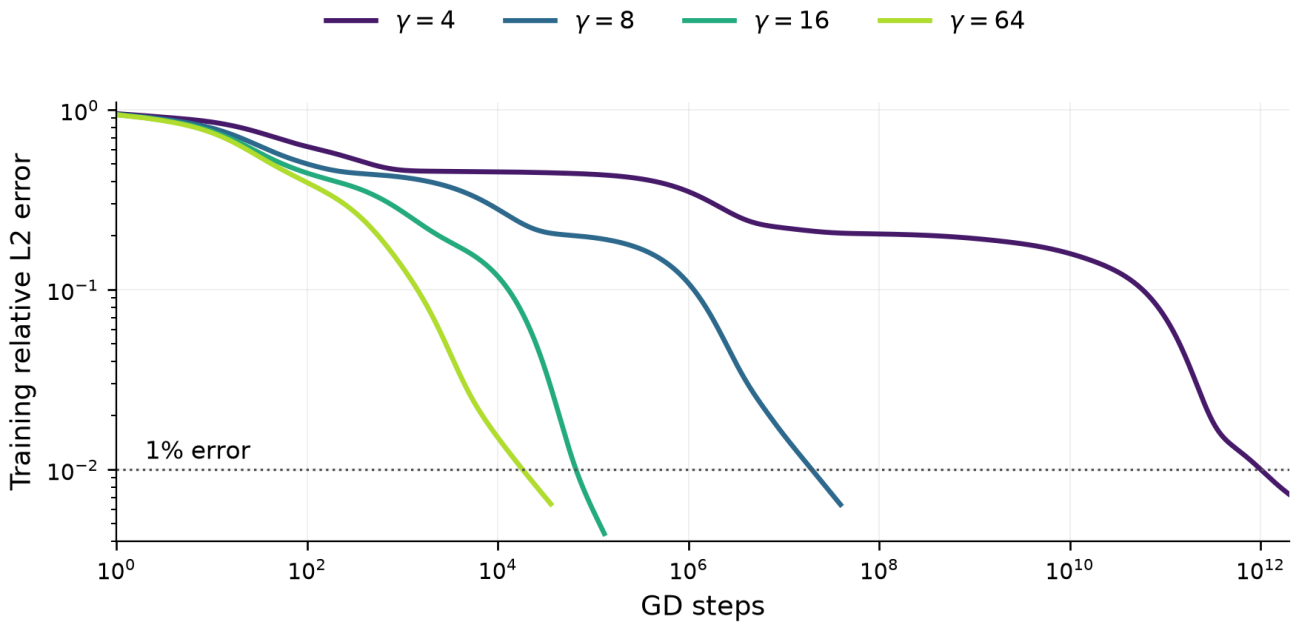


Figure 5. Full relative-error curves calculated from the finite-tanh spectrum and the center-integral spectrum, with 1% marked. The agreement extends over the approach to the threshold, rather than only at a fitted crossing. It does not certify dynamics near an unresolved machine-precision floor.

E.3. Numerical checks and limits

Small eigenvalues were computed from rectangular feature-matrix SVDs, squaring the singular values, rather than relying on an eigendecomposition of an explicitly rounded Gram matrix. An independent reconstruction using the alternative `gesdd` SVD, compared with the original `gesvd`, reproduced the finite-matrix step counts. Only the $\gamma = 4$ count changed, by 28 steps out of roughly 10^{12} . The explicit readout errors also agreed. These checks support the reported digits; they are not interval arithmetic.

The bound plots omit starting subspaces when either the relevant singular value or its cutoff gap is below $10 \epsilon_{\text{mach}} \max(m, W + 1) \sigma_1$. Additional checks screen unresolved compressed eigenvalues and compare the relative correction from whitening with a generalized eigenvalue calculation. The audit checked all 947 retained transition records against their directly computed ratios. Missing dashed segments indicate unresolved numerical evaluations, not evidence of a zero mathematical bound.

The explicit readouts in Figure 4 use a resolved, truncated SVD. Their directly evaluated errors demonstrate attainability at those levels; they are not exact representation floors. The slow band $(10^{-14}, 10^{-11}]$ was selected during the diagnostic, not before inspecting the spectrum. It supplies a valid band inequality once its rates and target energy are known, but the numerical values themselves are not directed-rounding certificates.

The target-weight control pairs components by eigenvalue rank and uses $\gamma_0 = 8$. It is a diagnostic of equation (3), not a unique causal decomposition: eigenvectors can rotate, and weights at nearly repeated or unresolved eigenvalues can depend on basis choices. Its conclusion is restricted to the observed improvement at 1% in this geometry.

E.4. What is proved and what is measured

Item	Status
GD recurrence and target-weighted error formula	Exact for fixed features, zero initial readout, and the stated updates.
Center-integral decomposition and Fourier finite-change identity	Exact under uniform center spacing; all corrections retained.
Finite-network eigenvalue-ratio lower bound	Proved inequality; useful improvement requires its right side to exceed the old ratio.
Positive-band necessary-step bound	Proved inequality; application requires the band's rates and target energy.
Corrections, ratio gains, and step counts in the figures	Floating-point evaluations, checked numerically but not interval certified.
Integral prediction within 0.3% at four slopes	Observed agreement in the specified experiment, not a uniform error theorem.

The current result does not provide a gamma-only asymptotic law for every finite-network ratio. That would require analytical estimates on $\lambda_{\min}(C_i)$, ϵ_i , and the normalization across a chosen family of geometries. The evidence here instead combines an explicit positive Fourier contribution, finite-matrix lower bounds that retain the corrections, and target-specific delays calculated without executing long training runs.

Appendix F. Source data and reproduction

This note follows the center-integral decomposition and finite-ratio calculations in `gamma_solve/spectrum_mechanism`. Its step predictor uses the finite center integral, not the separate collaborator notes' periodic construction. Earlier polynomial-tail bounds and scale-parameter-gradient experiments are not used in this argument.

The [experiment source](#) and [saved results](#) contain the calculations. The [figure manifest](#) maps each figure to its data files, and [figure_data.json](#) records the independent numerical audit. The rank-32 table uses reference slopes 8 and 16 in its respective rows; these are separate comparisons.